MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

Model Question Paper

MG1DSCICH100– Fundamentals of Industrial Chemistry

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions Answer **All** Questions

Each question carries **1** mark

1. Which of the following processes is an example of a chemical unit process? [Understand] [CO 1]

a) Distillation b) Evaporation c) Hydrogenation d) Filtration

2. How can the selectivity of a chemical reaction be improved? [Analyze] [CO 1, 2]

a) By increasing the temperature b) By decreasing the pressure c) By using a more selective catalyst d) All of the above

3. The octane number of iso-octane is [Remember] [CO 2, 4]

a) 100 b) 90 c) 50 d) 0

4. Which property of mica allows it to be split into thin sheets? [Apply] [CO 1, 2, 3]

a) High density b) Electrical conductivity c) Layered structure with van der Waals forces d) Low thermal conductivity

5) What is the primary ore of iron? [Remember] [CO 3]

a) Bauxite b) Hematite c) Galena d) Malachite

6) Which of the following is an example of wet corrosion? [Understand] [CO 1]

a) Corrosion of metal in the water b) Corrosion of iron in calcium Chloride c) Corrosion of titanium in dry chlorine d) Corrosion due to furnace gases

7) Quantum dots are [Understand] [CO 1]

a) Zero- Dimensional b) One- Dimensional c) Two- Dimensional d) Three- Dimensional

8) Which of the following is an example of the 'top-down' approach? [Understand] [CO 5]

a) Sol-gel synthesis b) Ball milling c) Chemical vapor deposition d) Self-assembly

9) What is the primary characteristic of nanomaterials? [Remember] [CO 5]

a) Size greater than 1 micron b) Size between 1 and 100 nanometers c) Size smaller than 1 millimeter d) Size between 100 and 1000 nanometers

10) What makes nanocomposites advantageous over conventional composites in industrial applications? [Apply] [CO 1, 2, 6]

a) Enhanced thermal stability and mechanical properties b) Easier manufacturing process

c) Lower cost d) Improved color properties

[1 x 10 = 10]

Part **B**

Short Answer Type QuestionsAnswer any **four** Questions

Each question carries **3** marks

11) Explain why safety measures are important in a chemical manufacturing plant. [Understand] [CO 1, 2]

12) Differentiate between unit processes and unit operations in chemical processing [Understand] [CO 1]

13) Distinguish between Chemical (Dry) corrosion and Electrochemical (wet) corrosion. [Understand] [CO1]

14) What is the primary chemical structure of cellulose? [Remember] [CO 1, 2, 4]

15) What is Gecko effect? How can Gecko effect be used to design advanced adhesives for industrial applications? [Understand] [CO 5]

16) Write any two applications of fullerenes. [Apply] [CO 1,2]

[3 x 4 = 12]

Part **C**

Short Essay Type Questions

Answer **four** Questions

Each question carries **7** marks

17) Describe the chemical composition of petroleum and natural gas. Explain how the different fractions of crude oil are used in everyday life. [Analyze] [CO 2, 4]

18) Analyse the advantages and disadvantages of batch and continuous processes in terms of efficiency, cost, and product quality. [Analyze] [CO 1, 2]

19) Compare the structures of graphite and diamond and explain how these structures contribute to their contrasting physical properties. [Understand] [CO 1, 2, 3]

20) Discuss the methods used for the extraction of copper from its ores. [Understand] [CO1, 3]

21) Explain the important properties and applications of Carbon Nano Tubes (CNTs). [Understand] [CO 1, 2]

22) Describe the sol-gel process in detail, outlining each step from sol formation to final material synthesis. [Understand] [CO 5]

[7 x 4 = 28]